

GRAPH NEURAL NETWORKS

AI506 — Advanced Machine Learning

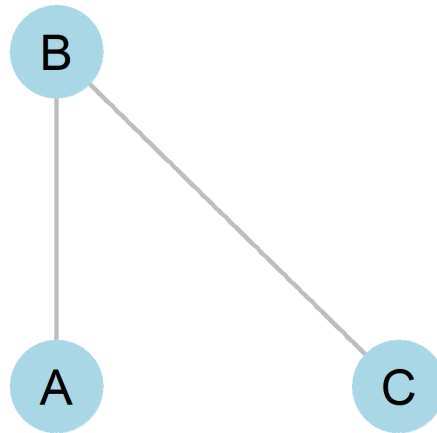
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May 2026

- **Graphs** as a data structure: **nodes, edges, neighborhoods, permutation invariance**
- **Message passing framework**
- Graph **convolution** and graph **attention**
- Practical considerations: **over-smoothing, jumping knowledge**

Let

nodes = $\{A, B, C\}$, edges = $\{(A, B), (B, C)\}$



$$h_i^{(l+1)} = \sigma\left(\text{UPD}\left(h_i^{(l)}, m_i^{(l)}\right)\right)$$

$$\text{UPD}\left(h_i^{(l)}, m_i^{(l)}\right) = W_h \cdot h_i^{(l)} + W_{\mathcal{N}} m_i^{(l)} + b^{(l)}$$

where

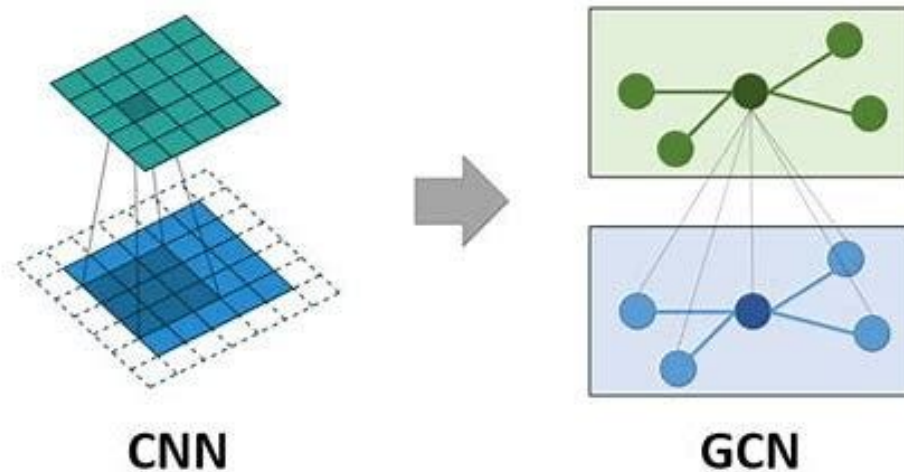
$$m_i = \text{AGG}_{j \in \mathcal{N}(i)} g\left(h_i^{(l)}, h_j^{(l)}\right)$$

g = some transformation of h_i and h_j

$$\text{AGG}_{\text{sum}} = \sum_{j \in \mathcal{N}(i)} h_j^{(l)}$$

$$\text{AGG}_{\text{avg}} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} h_j^{(l)}$$

$$\text{AGG}_{\text{max}} = \max_{j \in \mathcal{N}(i)} \left(h_j^{(l)} \right)$$



$$\text{AGG} = \sum_{j \in i} \frac{h_j}{\sqrt{\|\mathcal{N}(i)\| \cdot \|\mathcal{N}(j)\|}}$$

- Edge attributes (weight)

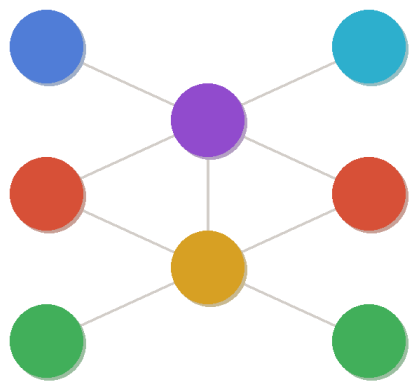
$$g\left(h_i^{(l)}, h_j^{(l)}\right) = a_{ij} \cdot h_j^{(l)}$$

$$\text{where } a_{ij} = \text{softmax}_{\mathcal{N}(i)}\left(a^T [Wh_i \parallel Wh_j]\right)$$

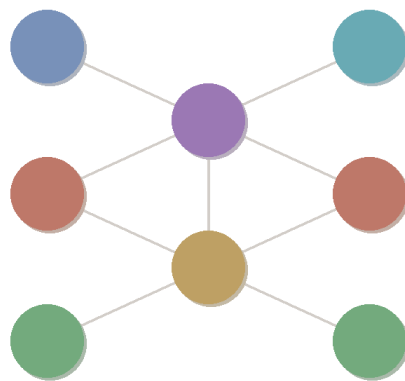
Where a is a learnable vector.

$$\text{AGG}_{\text{sum}} = \sum_{j \in \mathcal{N}(i)} a_{ij} h_j^{(l)}$$

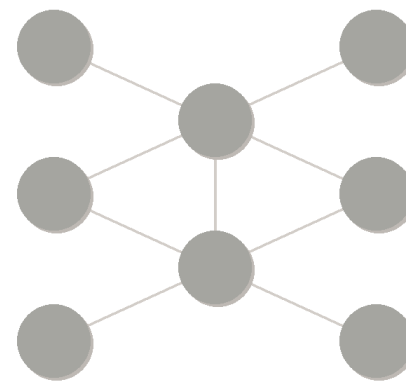
$$\text{AGG}_{\text{avg}} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} h_j^{(l+1)} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} \left(\frac{1}{\|\mathcal{N}(j)\|} \sum_{k \in \mathcal{N}(j)} h_k^{(l)} \right)$$



Layer 0



Layer 1



Layer 2

Concatenation:

$$z_i = \text{MLP}\left(h_i^{(1)} \# h_i^{(2)} \# \dots \# h_i^{(k)}\right)$$

Max pooling:

$$z_i = \text{MLP}(h_i^{\max}), \quad h_i^{\max} = \text{MaxPool}\left(h_i^{(1)}, h_i^{(2)}, \dots, h_i^{(k)}\right)$$

LSTM:

$$z_i = \text{MLP}(h_i^{\text{LSTM}}), \quad h_i^{\text{LSTM}} = \text{LSTM}\left(h_i^{(1)}, h_i^{(2)}, \dots, h_i^{(k)}\right)$$